HSV Split-and-Merge Fruit Segmentation for Controlled and Cross-Dataset Fruit Images

Computer Vision Assignment 2

Deggendorf Institute of Technology

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Abstract

This report presents a classical fruit-segmentation pipeline based on HSV colour statistics and a split-and-merge region framework. The system is trained on Fruits-360 training images, evaluated quantitatively on the Fruits-360 Test split, and demonstrated qualitatively on `test-multiple_fruits` scenes and on Fruits-262. The implemented pipeline combines guard-masked HSV statistics, quadtree splitting, region-adjacency merging, z-scored nearest-neighbour classification, and morphological post-processing. On the current checked-in evaluation reports, the system reaches 100.0% accuracy for the 3-class setup, 100.0% for the 5-class setup, and 99.3% for the 10-class setup on Fruits-360 Test. The strongest limitations appear during transfer to real-world imagery, where lighting, clutter, and hue overlap make region formation and classification less stable.

1. Introduction

Fruit segmentation is a good testbed for classical computer vision because fruit surfaces are strongly colour-driven, yet real scenes still contain shadows, reflections, textured backgrounds, and touching objects. This assignment requires a region-based method in HSV space, not a learned end-to-end detector. The chosen solution therefore emphasizes interpretability: each stage can be explained in terms of image statistics and geometric region operations.

The project follows the assignment constraints closely. HSV conversion may use a library call, but the actual segmentation logic is implemented in our own code. The split stage starts from at least 16 regions, homogeneity is judged from hue and saturation statistics, and evaluation is kept separate from tuning.

2. Datasets and Evaluation Protocol

Three datasets are used in different roles:

• **Fruits-360 Training**: used to build class references and tune parameters.

• **Fruits-360 Test**: used for quantitative evaluation through confusion matrices and per-class metrics.

• **Fruits-262** and `test-multiple_fruits`: used only as qualitative transfer tests.

This separation matters. Fruits-360 images are highly controlled: fruit instances are centered, backgrounds are nearly white, and object boundaries are clean. Fruits-262 is much more realistic, with leaves, clutter, natural light, shadows, and viewpoint changes. Good performance on Fruits-360 therefore does not guarantee good transfer to Fruits-262.

3. Class Selection

The current implementation uses the following ordered class list:

Rank	Fruits-360 folder	Display name
1	Cherry 1	Cherry
2	Orange 1	Orange
3	Banana 1	Banana
4	Avocado 1	Avocado
5	Cucumber 1	Cucumber
6	Cherry Wax Black 1	Cherry Black
7	Cucumber 3	Cucumber 3
8	Huckleberry 1	Huckleberry
9	Raspberry 1	Raspberry
10	Lychee 1	Lychee

The first three classes form the basic assignment setup. The first five extend the hue range into yellow-green and green, and the full ten-class configuration adds more difficult classes that occupy tighter colour neighborhoods.

4. Methodology

4.1 HSV Representation and Guard Mask

RGB mixes chromatic content with brightness, which makes it less stable under lighting changes. HSV separates hue, saturation, and value more explicitly, so it is easier to reason about colour similarity there. However, hue becomes unstable for dark or weakly saturated pixels. The pipeline therefore uses a guard mask that excludes pixels below minimum value and saturation thresholds from downstream hue statistics.

4.2 Preprocessing

Before segmentation, the image is smoothed with a median filter followed by a light Gaussian low-pass filter. The purpose is not to blur the fruit boundaries, but to suppress local noise and tiny specular variations that would otherwise make the split criterion fire too aggressively.

4.3 Split Phase

The split stage uses a quadtree decomposition. The image is subdivided until each region is sufficiently homogeneous according to hue and saturation statistics, or until a minimum size is reached. The assignment requirement of at least 16 initial regions is satisfied by forcing a sufficiently deep initial subdivision.

4.4 Merge Phase

After splitting, many neighboring blocks still belong to the same fruit. A region-adjacency graph is therefore used to merge spatially adjacent regions when their colour statistics remain compatible. A Sobel-derived edge cue is used as a merge veto so that sharp object boundaries are less likely to collapse.

4.5 Region Classification

Each merged region is described by a compact HSV feature vector. Class references are built from Fruits-360 Training images, and each region is assigned to the nearest class in a z-scored feature space. Regions too far from all references are rejected as background instead of being forced into an incorrect fruit class.

4.6 Post-processing

Per-class masks are refined through morphological cleanup and area filtering. This step removes tiny fragments and fills small holes so the final overlay is easier to interpret visually.

5. Pipeline Illustration

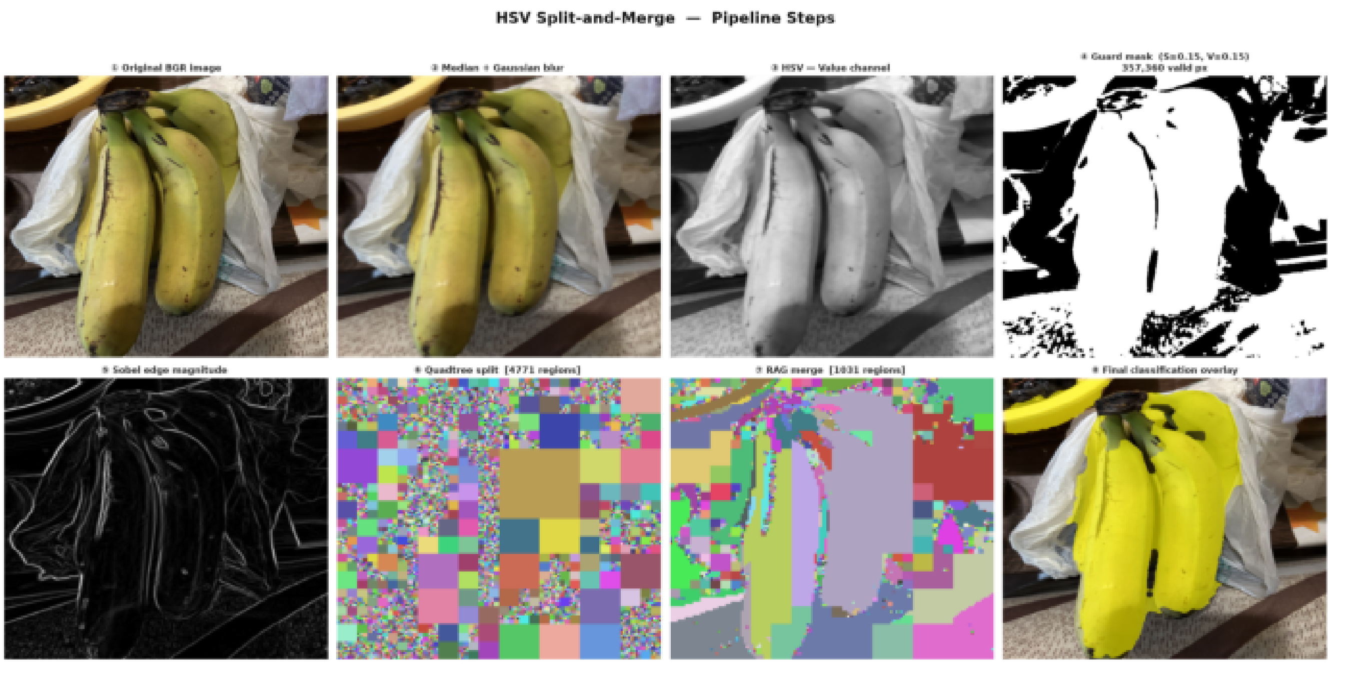

Figure: Pipeline steps

Figure 1 shows the major intermediate stages used in the final implementation. The guard mask, region decomposition, merge result, and final overlay illustrate how the pipeline transforms a scene image from low-level colour statistics into class-labelled fruit regions.

6. Quantitative Results on Fruits-360 Test

The current checked-in evaluation reports show:

Configuration	Classes	Accuracy
3-class	Cherry, Orange, Banana	100.0%
5-class	Cherry, Orange, Banana, Avocado, Cucumber	100.0%

10-class	Full set	99.3%

6.1 3-Class Result

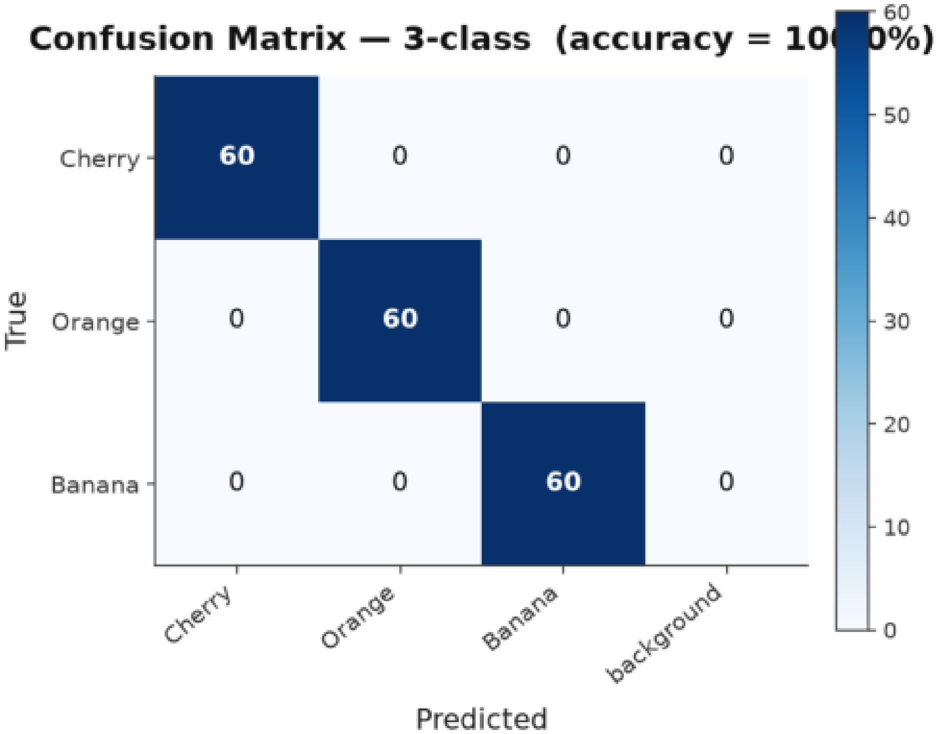

Figure: 3-class confusion matrix

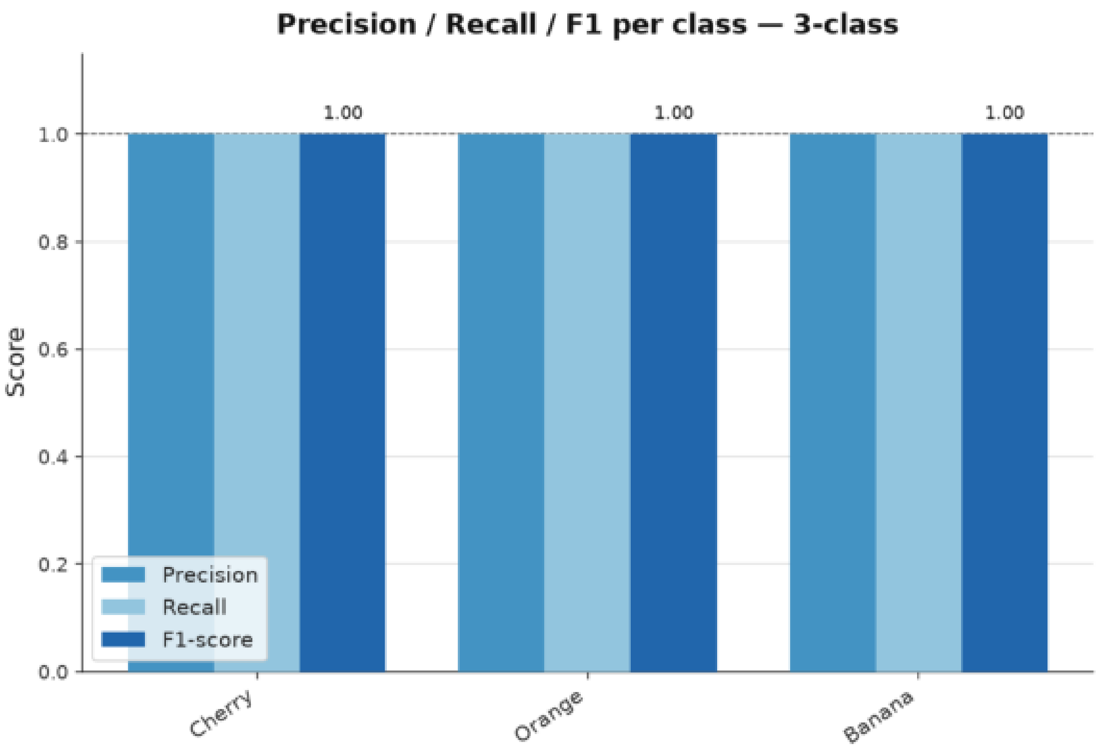

Figure: 3-class metrics

The 3-class setting is the most stable because Cherry, Orange, and Banana occupy well-separated regions of HSV space in the controlled Fruits-360 images.

6.2 5-Class Result

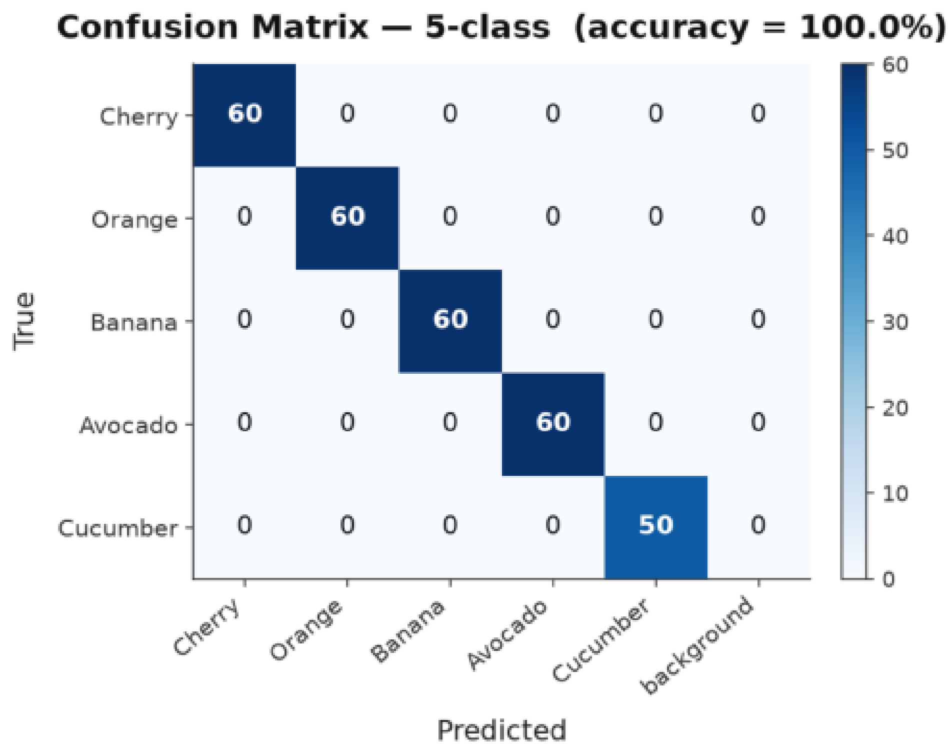

Figure: 5-class confusion matrix

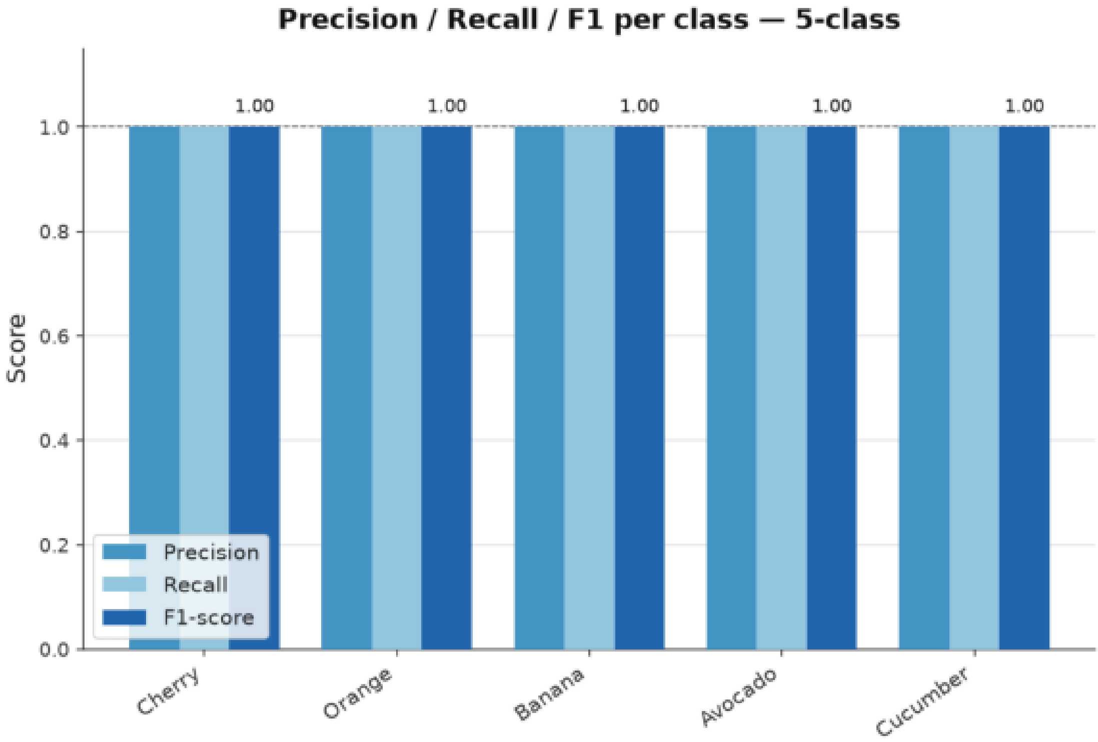

Figure: 5-class metrics

The 5-class setup remains perfect on the current evaluation set. Adding Avocado and Cucumber increases complexity, but the controlled background and relatively clean hue separation still keep errors low.

6.3 10-Class Result

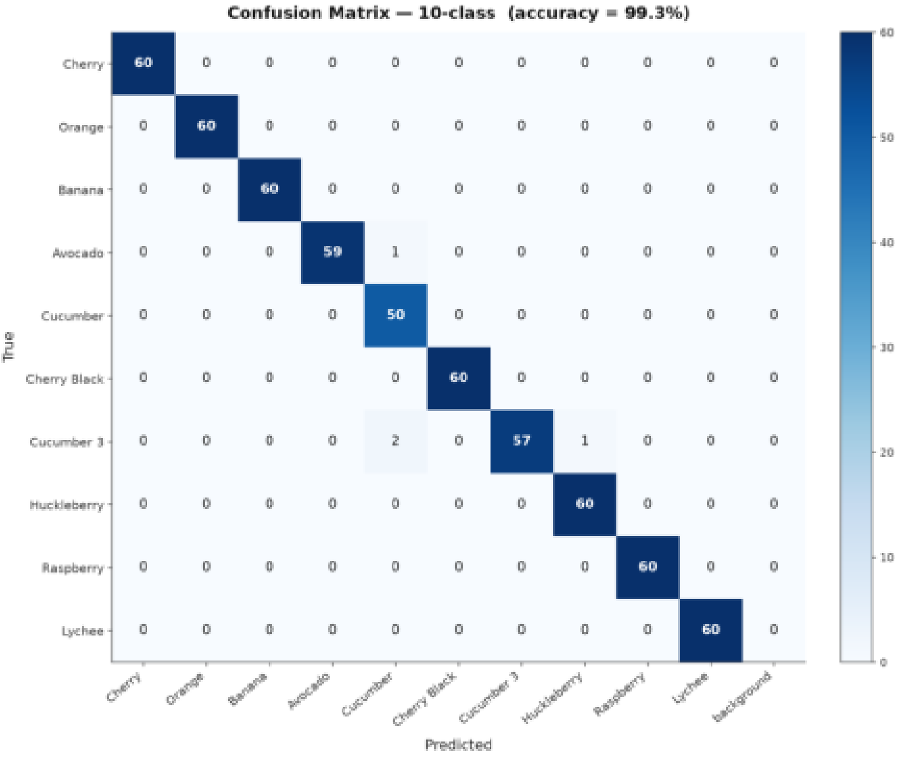

Figure: 10-class confusion matrix

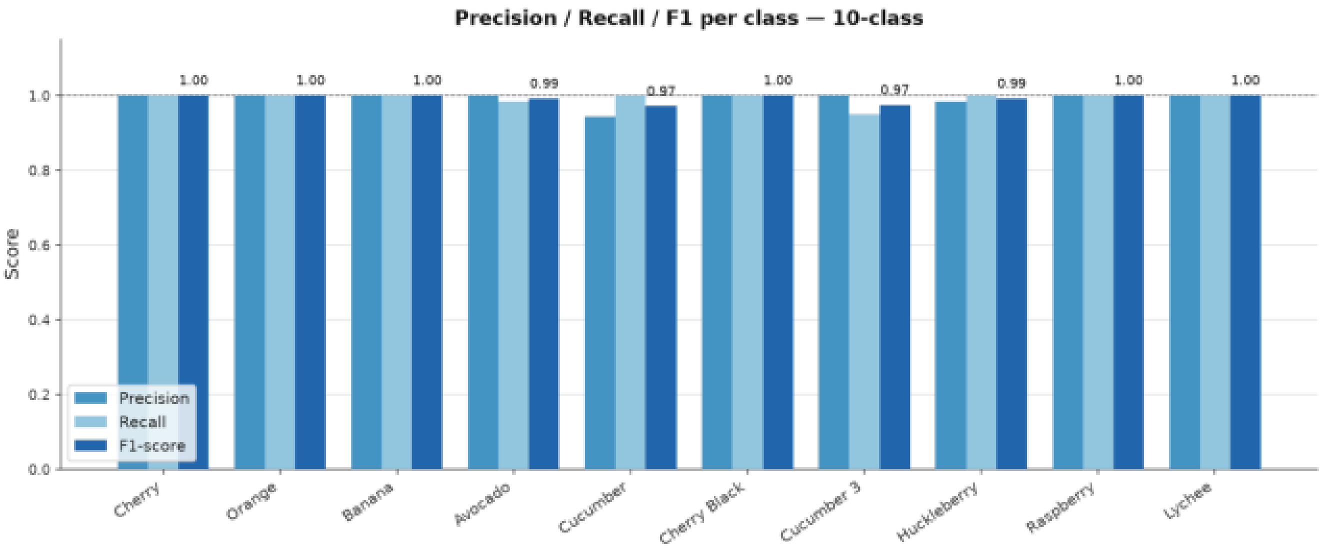

Figure: 10-class metrics

The 10-class setting reaches 99.3% accuracy. Most errors occur at known colour boundary cases, especially where green, cyan, dark red, or low-saturation fruit classes become visually close in the chosen feature space.

7. Qualitative Transfer Results

7.1 Fruits-262 Class Selection

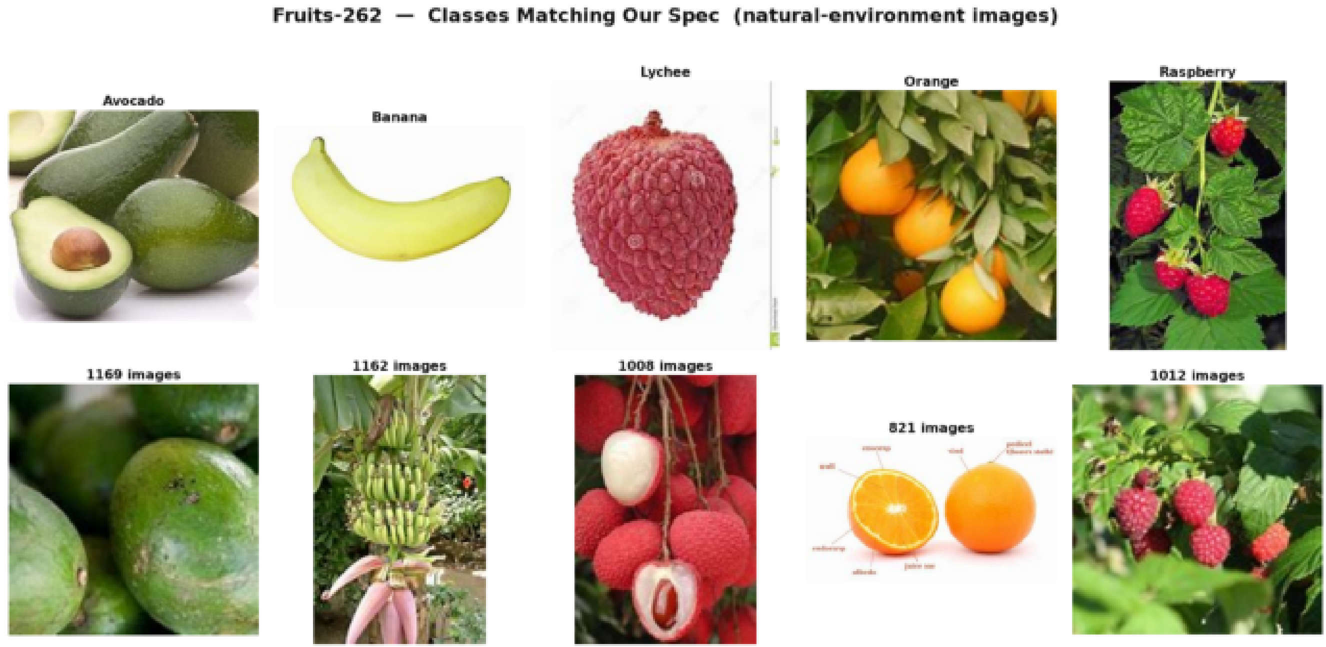

Figure: Fruits-262 matching classes

Fruits-262 provides a much more realistic test of robustness because the fruit is no longer presented on a clean white background. Leaves, natural lighting, and non-fruit objects make both segmentation and classification harder.

7.2 3-Class Cross-Dataset Proof

Figure: Fruits-262 3-class proof

The 3-class proof-of-concept shows that the model can transfer basic colour-based knowledge from Fruits-360 to more realistic photos. Cherry, Orange, and Banana remain distinguishable in many cases because their colour separation is strong.

7.3 Composite Cross-Dataset Scene

Figure: Fruits-262 3-class scene

This scene demonstrates the practical challenge of cross-dataset transfer. Even when the expected fruits are present, their segmented shapes can degrade because the background introduces competing colours and weak boundaries.

7.4 10-Class Multi-Fruit Overlay

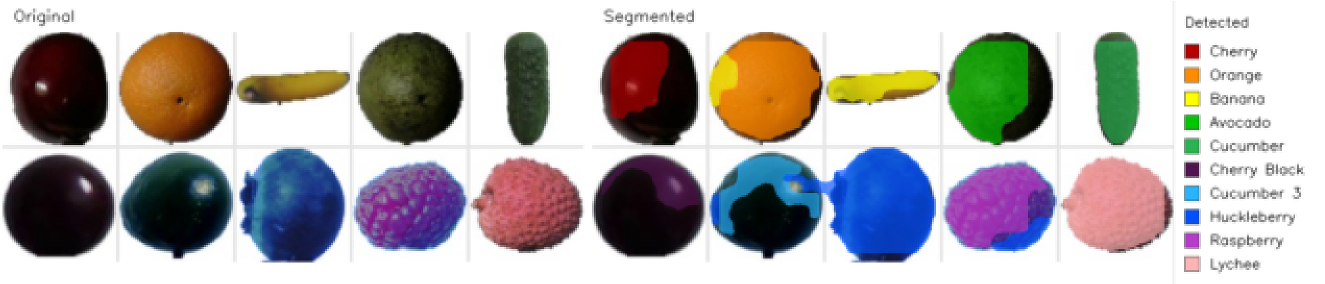

Figure: 10-class multi-fruit overlay

The 10-class composite overlay is useful for presentation because it shows the full legend and demonstrates that the system can keep many fruit classes distinct in one visualization when the visual conditions are favorable.

8. Limitations and Drawbacks

The pipeline has several important limitations:

• **Hue overlap** is still the dominant failure mode. Different fruits can occupy similar HSV regions, especially under changing illumination.

• **Blocky boundaries** are inherited from the quadtree split representation.

• **Threshold sensitivity** remains significant. A parameter set that works well on Fruits-360 may not transfer equally well to natural scenes.

• **Dark or reflective regions** may be rejected by the guard mask, causing under-segmentation or background labelling.

• **Colour-only reasoning** is insufficient when non-fruit objects share similar colour distributions.

• **Closed-set behavior** means unknown fruits may still be mapped to the nearest known class if the rejection logic is not strict enough.

These drawbacks are not just implementation details; they follow from the choice of a classical colour-driven segmentation strategy.

9. Conclusion

The final system satisfies the assignment's core requirements with a fully interpretable HSV split-and-merge pipeline. It performs extremely well on the controlled Fruits-360 Test data and produces clear qualitative overlays for both multi-fruit scenes and cross-dataset examples. The main lesson is that strong in-dataset accuracy is achievable with classical methods, but robust transfer to real-world scenes remains limited by colour ambiguity, threshold sensitivity, and scene complexity.

References

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